

Machine Learning (MC03094121)

UNIT-1 : Introduction to Machine Learning

Topic 1 : Introduction to Machine Learning

Definition

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and improve their performance without being explicitly programmed.

Instead of writing rules manually, the machine learns from examples.

Definition (Exam)

Machine Learning is a field of Artificial Intelligence that allows computers to learn from data and make predictions or decisions without being explicitly programmed.

Traditional Programming vs Machine Learning

Traditional Programming

Programmer writes rules

Input + Program $\rightarrow$ Output

Static

Suitable for fixed problems

Machine Learning

Machine learns rules automatically

Input + Output $\rightarrow$ Model

Learns continuously

Suitable for complex problems

Real-Life Examples

- Gmail Spam Detection
 - Netflix Movie Recommendation
 - Amazon Product Recommendation
 - Google Translate
 - Face Unlock
 - Speech Recognition
 - Chatbots
 - Fraud Detection
 - Self-driving Cars
-

Why Machine Learning?

Traditional programming cannot solve problems where rules are too complex.

Example:

Can we manually write rules to identify every cat image?

Impossible.

Machine Learning learns these patterns automatically from thousands of images.

Characteristics of Machine Learning

- Learns from data
 - Improves with experience
 - Makes predictions
 - Finds hidden patterns
 - Handles large datasets
 - Automates decision making
-

Advantages

- High accuracy
 - Automation
 - Fast prediction
 - Handles big data
 - Continuous improvement
 - Reduces human effort
-

Limitations

- Requires large datasets
 - High computational cost
 - Training takes time
 - Data quality affects performance
 - Difficult to explain some models
-

Applications

- Healthcare
 - Banking
 - Education
 - Agriculture
 - Finance
 - Cyber Security
 - Marketing
 - Manufacturing
 - Entertainment
 - Transportation
-

Example

A bank has data of 10,000 previous customers.

Machine Learning studies:

- Income
- Age
- Loan Amount
- Previous Loans

Then predicts whether a new customer will repay the loan.

Topic 2 : Human Learning vs Machine Learning

Human Learning

Humans learn through:

- Observation
- Experience
- Practice
- Mistakes
- Memory

Example:

A child touches fire once and learns not to touch it again.

Machine Learning

Machines learn using:

- Historical Data
- Training Algorithms
- Mathematical Models
- Pattern Recognition

Example:

A spam filter studies thousands of emails before identifying spam.

Comparison

Human Learning	Machine Learning
Uses brain	Uses algorithms
Learns through experience	Learns through data
Can think logically	Cannot think like humans
Uses emotions	No emotions
Learns slowly	Learns quickly with sufficient data
Can generalize naturally	Depends on training data

Similarities

- Both improve with experience.
- Both recognize patterns.
- Both make decisions based on previous knowledge.
- Both require practice/training.

Differences

Human	Machine
Biological intelligence	Artificial intelligence
Learns naturally	Learns mathematically
Can imagine	Cannot imagine
Uses reasoning	Uses probability
Makes emotional decisions	Makes data-driven decisions

Topic 3 : Types of Machine Learning

Machine Learning is mainly divided into four types.

1. Supervised Learning

Definition

The model learns from **labeled data** (input + correct output).

Goal:

Predict the output for new data.

Examples

- House Price Prediction
- Email Spam Detection
- Student Result Prediction

Algorithms

- Linear Regression
 - Decision Tree
 - k-NN
 - Random Forest
 - SVM
-

2. Unsupervised Learning

Definition

The model learns from **unlabeled data**.

Goal:

Find hidden patterns or groups.

Examples

- Customer Segmentation
- Market Basket Analysis
- Product Recommendation

Algorithms

- K-Means
 - Hierarchical Clustering
 - DBSCAN
 - Apriori
-

3. Semi-Supervised Learning

Uses both labeled and unlabeled data.

Example:

Only 500 images are labeled, but 10,000 images are available.

4. Reinforcement Learning

The model learns by interacting with the environment using rewards and penalties.

Example:

- Robot Navigation
 - Chess Playing
 - Self-driving Cars
 - Game Playing (AlphaGo)
-

Comparison Table

Type	Data	Goal
Supervised	Labeled	Prediction
Unsupervised	Unlabeled	Pattern Finding
Semi-supervised	Mixed	Better Accuracy
Reinforcement	Reward Based	Decision Making

Topic 4 : Applications of Machine Learning

Machine Learning is widely used in many domains.

Healthcare

- Disease prediction
- Medical image analysis
- Drug discovery

Banking

- Fraud detection
- Credit scoring
- Loan approval

E-commerce

- Product recommendation
- Customer segmentation
- Demand forecasting

Education

- Personalized learning
- Student performance prediction
- Intelligent tutoring systems

Agriculture

- Crop disease detection
- Yield prediction
- Smart irrigation

Cyber Security

- Intrusion detection
- Malware detection
- Spam filtering

Transportation

- Self-driving cars
- Traffic prediction
- Route optimization

Entertainment

- Movie recommendations
- Music recommendations
- Content personalization

Topic 5: Tools for Machine Learning

What are Machine Learning Tools?

Machine Learning tools are software, programming languages, libraries, and platforms used to develop, train, test, evaluate, and deploy Machine Learning models.

These tools simplify complex ML tasks and improve development speed.

Why are ML Tools Required?

Machine Learning tools help us to:

- Collect and preprocess data
- Build ML models
- Train algorithms
- Visualize data
- Evaluate model performance
- Deploy models into real applications

Without these tools, implementing Machine Learning would be very difficult and time-consuming.

Popular Machine Learning Tools

1. Python

Python is the most popular programming language for Machine Learning.

Features

- Easy syntax
- Open source
- Large community support
- Huge collection of ML libraries

Applications

- Data Analysis
- Artificial Intelligence
- Deep Learning
- Data Science
- Automation

Advantages

- Beginner-friendly
 - Cross-platform
 - Extensive library support
-

2. Jupyter Notebook

Jupyter Notebook is an interactive environment used for writing and executing Python code.

Features

- Execute code cell by cell
- Combine code, text, images, and graphs
- Interactive output

Advantages

- Easy experimentation
 - Good for teaching and research
 - Widely used by Data Scientists
-

3. Google Colab

Google Colaboratory (Colab) is a cloud-based Jupyter Notebook provided by Google.

Features

- No installation required
- Free GPU and TPU support
- Accessible from any browser
- Auto-saving on Google Drive

Advantages

- Easy collaboration
 - Cloud execution
 - Suitable for Deep Learning projects
-

4. NumPy

NumPy is a Python library used for numerical computing.

Features

- Fast array operations
- Matrix calculations
- Mathematical functions

Applications

- Linear Algebra
 - Statistics
 - Scientific Computing
-

5. Pandas

Pandas is a Python library used for data manipulation and analysis.

Features

- DataFrame
- Reading CSV and Excel files
- Filtering and sorting
- Handling missing values

Applications

- Data Cleaning
 - Data Analysis
 - Feature Engineering
-

6. Matplotlib

Matplotlib is a data visualization library.

Features

- Line Chart
- Bar Chart
- Pie Chart
- Histogram
- Scatter Plot

Applications

- Data visualization
- Trend analysis
- Graph plotting

7. Scikit-learn

Scikit-learn is the most popular Machine Learning library in Python.

Features

Supports

- Classification
- Regression
- Clustering
- Feature Selection
- Model Evaluation

Common Algorithms

- Linear Regression
- KNN
- Decision Tree
- Naive Bayes
- K-Means

8. TensorFlow

TensorFlow is an open-source framework developed by Google.

Applications

- Deep Learning
- Neural Networks
- Image Recognition
- Speech Recognition

9. Keras

Keras is a high-level Deep Learning API that works on top of TensorFlow.

Features

- Easy to learn
- Less coding
- Fast model development

Comparison of Popular ML Tools

Tool	Purpose
Python	Programming Language
Jupyter Notebook	Interactive Coding Environment
Google Colab	Cloud-based Notebook
NumPy	Numerical Computation
Pandas	Data Manipulation
Matplotlib	Data Visualization
Scikit-learn	Machine Learning Algorithms
TensorFlow	Deep Learning Framework
Keras	Neural Network Development

Real-Life Example

Suppose we want to predict student performance.

Workflow:

Student Data (CSV)

↓

Pandas → Read Dataset

↓

NumPy → Numerical Operations

↓

Matplotlib → Visualize Data

↓

Scikit-learn → Train Decision Tree Model

↓

Evaluate Accuracy

↓

Deploy Model

Advantages of ML Tools

- Reduce coding effort
- Faster model development
- Easy visualization
- High computational efficiency
- Open-source ecosystem
- Industry standard tools

Disadvantages

- Some tools require high system resources
- Beginners need time to learn multiple libraries
- Version compatibility issues may occur
- Large datasets require more memory

Topic 6: Machine Learning Activities

What are Machine Learning Activities?

Machine Learning Activities are the sequence of steps followed to build an effective Machine Learning model. These activities convert raw data into useful predictions or decisions.

Steps in Machine Learning Activities

Step 1: Problem Definition

Clearly identify the problem to be solved.

Examples:

- Predict house prices
- Detect spam emails
- Recognize handwritten digits

Step 2: Data Collection

Collect relevant data from various sources such as:

- Databases
- CSV/Excel files
- APIs
- Sensors
- Websites

Example:

Customer purchase history for predicting future sales.

Step 3: Data Pre-processing

Clean and prepare the data.

Tasks include:

- Removing missing values
- Removing duplicates
- Handling outliers
- Encoding categorical data
- Feature scaling

Result:

A clean dataset ready for training.

Step 4: Feature Engineering

Select or create useful features that improve prediction accuracy.

Example:

Original Feature:

Date of Birth

New Feature:

Age

Step 5: Model Selection

Choose an appropriate Machine Learning algorithm.

Examples:

Classification → Decision Tree, Naive Bayes, KNN

Regression → Linear Regression

Clustering → K-Means

Step 6: Model Training

The algorithm learns patterns from training data.

Training Data → ML Algorithm → Trained Model

Step 7: Model Evaluation

Evaluate model performance using testing data.

Common evaluation metrics:

- Accuracy
 - Precision
 - Recall
 - F1-Score
 - RMSE (Regression)
-

Step 8: Model Improvement

Improve performance by:

- Hyperparameter tuning
 - Feature selection
 - Collecting more data
 - Cross Validation
-

Step 9: Model Deployment

Deploy the trained model into real-world applications.

Examples:

- Banking fraud detection
 - Netflix recommendation
 - Face recognition
-

Complete Machine Learning Workflow

Problem Definition



Data Collection



Data Pre-processing



Feature Engineering



Model Selection



Model Training



Model Evaluation



Model Improvement



Deployment



Prediction

Advantages

- Systematic development process
 - Higher prediction accuracy
 - Better model reliability
 - Easy maintenance
-

Topic 7: Data Structures for Machine Learning

Introduction

Data Structures are specialized ways of organizing and storing data so that Machine Learning algorithms can process it efficiently.

Python provides many built-in data structures for ML.

Common Data Structures

1. List

Stores multiple values in sequence.

Example:

```
marks = [78, 82, 91, 67]
```

Uses:

- Store datasets
 - Training samples
 - Feature values
-

2. Tuple

An ordered collection that cannot be modified.

Example:

```
point = (10, 20)
```

Uses:

- Fixed records
 - Coordinates
-

3. Dictionary

Stores data as Key : Value pairs.

Example:

```
student = {  
    "Name": "Rahul",  
    "Age": 20  
}
```

Uses:

- Label mapping
 - Feature storage
 - Dataset information
-

4. Set

Stores unique values only.

Example:

```
cities = {"Rajkot", "Ahmedabad", "Surat"}
```

Uses:

- Remove duplicates
 - Unique categories
-

5. NumPy Array

Special array for numerical computation.

Example:

```
import numpy as np  
  
a = np.array([10, 20, 30])
```

Advantages:

- Faster than Python List
 - Less memory
 - Mathematical operations
-

6. Pandas DataFrame

Most widely used data structure in Machine Learning.

Example:

```
import pandas as pd

df = pd.read_csv("student.csv")
```

Advantages:

- Stores data in rows and columns
- Easy filtering
- Easy analysis
- Missing value handling

Comparison

Data Structure	Ordered Mutable		ML Usage
List	Yes	Yes	Dataset storage
Tuple	Yes	No	Fixed records
Dictionary	Yes	Yes	Labels & Features
Set	No	Yes	Unique values
NumPy Array	Yes	Yes	Numerical computation
Pandas DataFrame	Yes	Yes	Dataset processing

Real-life Example

Titanic Dataset

Passenger Age Gender Fare Survived

John 22 Male 250 Yes

This dataset is stored in a **Pandas DataFrame**, which is then used for:

- Data cleaning
- Feature engineering
- Model training
- Prediction

Advantages

- Efficient data organization
- Faster computation
- Easy manipulation
- Better memory utilization

- Essential for Machine Learning libraries
-

UNIT-2: Preparing to Model & Feature Engineering

Topic 1: Data Pre-processing

What is Data Pre-processing?

Data Pre-processing is the process of converting raw, incomplete and inconsistent data into a clean and useful format before applying Machine Learning algorithms.

It is one of the most important steps because the quality of data directly affects model accuracy.

Formula

Good Data → Good Model

Bad Data → Poor Model

Why Data Pre-processing is Required?

Real-world datasets usually contain:

- Missing values
- Duplicate records
- Noise
- Incorrect values
- Different scales
- Categorical data

Without preprocessing, ML algorithms may produce inaccurate predictions.

Steps of Data Pre-processing

Step 1: Collect Data

Gather data from

- Database
- CSV files
- Excel
- APIs
- Sensors
- Websites

Example:

Student Marks Dataset

Name Math Science

A 80 75

B 65 82

Step 2: Handle Missing Values

Sometimes some values are absent.

Example

Age Salary

25 30000

30 NULL

28 40000

Methods

- Delete rows
 - Replace with Mean
 - Replace with Median
 - Replace with Mode
-

Step 3: Remove Duplicate Data

Duplicate records increase bias.

Example

Raj 22

Raj 22

Keep only one record.

Step 4: Handle Outliers

Outliers are abnormal values.

Example

Student Marks

65

70

68

72

250 ← Outlier

Methods

- Remove
 - Replace
 - IQR Method
 - Z-score
-

Step 5: Feature Scaling

Different attributes may have different ranges.

Example

Age = 22

Salary = 500000

Salary dominates Age.

Scaling makes both equally important.

Methods

- Normalization
 - Standardization
-

Step 6: Encode Categorical Data

Machine Learning algorithms understand numbers only.

Example

Male $\rightarrow$ 0

Female $\rightarrow$ 1

Step 7: Split Dataset

Divide into

- Training Data
- Testing Data

Common Ratio

80% Training

20% Testing

Advantages

- Improves accuracy
 - Removes noise
 - Faster learning
 - Better prediction
 - Better visualization
-

Disadvantages

- Time consuming
 - Wrong preprocessing may reduce accuracy
 - Requires domain knowledge
-

Real-life Example

Online Shopping Website

Raw Data

Customer Name

Age

Gender

Income

Purchased

↓

Missing values removed

↓

Gender encoded

↓

Income normalized

↓

Train ML Model

↓

Predict Future Purchases

Topic 2: Selecting a Model

What is Model Selection?

Model Selection is the process of choosing the most suitable Machine Learning algorithm for solving a particular problem.

Different problems require different algorithms.

Types of Problems and Models

Classification Problems

Output is a category.

Examples

Spam or Not Spam

Disease Yes or No

Algorithms

- Decision Tree
 - Naive Bayes
 - KNN
 - SVM
-

Regression Problems

Output is continuous.

Examples

House Price

Salary Prediction

Algorithms

- Linear Regression
 - Multiple Regression
-

Clustering Problems

No output labels.

Algorithms

- K-Means
 - DBSCAN
 - Hierarchical Clustering
-

Factors Affecting Model Selection

Dataset Size

Small dataset

→ KNN

Large dataset

→ Decision Tree

Data Type

Numerical

Categorical

Mixed

Accuracy Requirement

Medical Diagnosis requires high accuracy.

Movie Recommendation may tolerate slightly lower accuracy.

Training Time

Simple algorithms

Fast

Complex algorithms

Slow

Interpretability

Some models are easy to understand.

Example

Decision Tree

Some models are difficult.

Example

Neural Networks

Model Selection Process

Problem



Collect Data



Preprocess Data



Choose Candidate Models



Train Models



Evaluate Models



Select Best Model

Example

Predict Student Result

Possible Models

Decision Tree

KNN

Naive Bayes

Train all three

Compare Accuracy

Select highest accuracy model.

Advantages

- Better prediction
 - Higher accuracy
 - Efficient learning
 - Reduced overfitting
-

Disadvantages

- Time consuming
 - Requires experimentation
 - No universal best algorithm
-

Topic 3: Training a Model

Definition

Model Training is the process in which a Machine Learning algorithm learns patterns and relationships from the training dataset to make predictions on new data.

During training, the algorithm adjusts its internal parameters to minimize prediction errors.

Definition (Exam)

Model Training is the process of feeding training data to a Machine Learning algorithm so that it learns the relationship between input and output data.

Need for Model Training

Model training is required because:

- It helps the model learn from historical data.

- It identifies hidden patterns and relationships.
 - It improves prediction accuracy.
 - It prepares the model for real-world data.
-

Steps in Model Training

Step 1: Prepare Dataset

Collect and preprocess the data.



Step 2: Split Dataset

- Training Data (70–80%)
- Testing Data (20–30%)



Step 3: Select Algorithm

Choose an appropriate algorithm such as:

- Decision Tree
- Linear Regression
- KNN
- SVM



Step 4: Train the Model

The algorithm learns from the training dataset.



Step 5: Generate Trained Model

The trained model stores learned patterns.



Step 6: Test the Model

Evaluate the model using testing data.

Example

Suppose we want to predict house prices.

Training Data:

Area Price

1000 25 Lakh

1200 30 Lakh

1500 38 Lakh

The algorithm studies this data and creates a prediction model.

Advantages

- Learns automatically from data.
 - Improves prediction accuracy.
 - Detects hidden relationships.
 - Reduces manual programming.
-

Disadvantages

- Requires quality data.
 - Training may take a long time.
 - Large datasets require more computational resources.
-

Topic 4: Model Representation and Interpretability

Definition

Model Representation refers to the way a Machine Learning model stores the relationship between input features and output predictions.

Interpretability refers to how easily a human can understand the model's predictions.

Importance

- Helps understand how the model works.
 - Builds user trust.
 - Makes debugging easier.
 - Supports decision making.
-

Types of Models

Interpretable Models

These models are easy to understand.

Examples:

- Linear Regression
 - Decision Tree
-

Non-Interpretable Models

These models are difficult to understand.

Examples:

- Neural Networks
- Deep Learning Models

These are often called **Black Box Models**.

Example

A Decision Tree clearly shows the decision path taken to classify data.

A Deep Neural Network gives accurate predictions but does not clearly explain how it reached the result.

Advantages

- Easy to explain results.
 - Increases transparency.
 - Helps identify model errors.
-

Limitations

- Highly interpretable models may be less accurate.
 - Complex models are difficult to explain.
-

Topic 5: Evaluating Performance of a Model

Definition

Model Evaluation is the process of measuring how well a trained Machine Learning model performs on unseen data.

Why Evaluation is Necessary?

- Measures prediction accuracy.
 - Compares different models.
 - Detects overfitting and underfitting.
 - Helps choose the best model.
-

Common Evaluation Metrics

1. Accuracy

Accuracy is the percentage of correctly predicted observations.

Formula

Accuracy = (Correct Predictions / Total Predictions) × 100

2. Precision

Precision measures how many predicted positive values are actually positive.

3. Recall

Recall measures how many actual positive values are correctly identified.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall.

5. Confusion Matrix

A Confusion Matrix is a table used to evaluate the performance of classification models.

It consists of:

- True Positive (TP)
 - True Negative (TN)
 - False Positive (FP)
 - False Negative (FN)
-

Advantages

- Measures model quality.
 - Helps compare models.
 - Improves prediction reliability.
-

Limitations

- Different metrics are suitable for different problems.
 - Accuracy alone may be misleading for imbalanced datasets.
-

Topic 6: Improving Performance of a Model

Definition

Improving model performance means increasing the prediction accuracy and reducing errors by optimizing the Machine Learning model.

Techniques to Improve Performance

1. Collect More Data

More quality data generally improves prediction accuracy.

2. Better Data Preprocessing

- Remove missing values.
 - Remove duplicates.
 - Handle outliers.
 - Normalize data.
-

3. Feature Engineering

Select important features and remove unnecessary ones.

4. Hyperparameter Tuning

Adjust parameters such as:

- Learning Rate
- Number of Neighbors (K)
- Tree Depth

to obtain better results.

5. Cross Validation

Divide the dataset into multiple parts and evaluate the model several times for better reliability.

6. Ensemble Methods

Combine multiple models to improve prediction accuracy.

Examples:

- Random Forest
- Gradient Boosting

7. Reduce Overfitting

Methods include:

- Regularization
- Pruning
- Dropout (for Neural Networks)

Advantages

- Higher prediction accuracy.
- Better generalization.
- More reliable model.
- Reduced prediction errors.

Limitations

- Increased computational cost.
 - Requires experimentation.
 - More training time.
-

UNIT–3 : Modelling and Evaluation

Topic 1: Introduction to Feature Engineering

Definition

Feature Engineering is the process of selecting, modifying, creating, or transforming input variables (features) from raw data so that Machine Learning models can learn more effectively and produce accurate predictions.

A **feature** is an individual measurable property or characteristic of the data.

Definition (Exam)

Feature Engineering is the process of extracting, selecting, and transforming useful features from raw data to improve the performance of a Machine Learning model.

Why Feature Engineering is Required?

Real-world data is often incomplete, inconsistent, or contains irrelevant information. Feature Engineering helps convert this raw data into meaningful features that improve the accuracy of Machine Learning models.

Objectives of Feature Engineering

- Improve model accuracy.
 - Reduce unnecessary data.
 - Speed up model training.
 - Improve prediction quality.
 - Reduce overfitting.
-

Steps in Feature Engineering

1. Collect raw data.
 2. Clean the data.
 3. Handle missing values.
 4. Transform features.
 5. Select important features.
 6. Train the Machine Learning model.
-

Example

Raw Dataset

Name	DOB	Salary
Rahul	10-05-2000	35000

After Feature Engineering

Age	Salary
-----	--------

Age Salary

25 35000

Here, **Age** is created from the **Date of Birth**, making the data more useful for the model.

Advantages

- Increases prediction accuracy.
 - Reduces training time.
 - Improves model performance.
 - Removes irrelevant information.
-

Disadvantages

- Time-consuming process.
 - Requires domain knowledge.
 - Wrong feature selection can reduce accuracy.
-

Topic 2: Feature Transformation

Definition

Feature Transformation is the process of converting existing features into a more suitable format so that Machine Learning algorithms can process them efficiently.

Why Feature Transformation is Required?

Different features may have different units and scales. Feature transformation ensures that all features are represented in a consistent format.

Common Feature Transformation Techniques

1. Normalization

Converts feature values into a fixed range (usually 0 to 1).

Example:

Salary: 25,000–1,00,000

↓

Normalized values: 0–1

2. Standardization

Transforms data so that it has:

- Mean = 0
 - Standard Deviation = 1
-

3. Encoding

Converts categorical values into numerical values.

Example

Male → 0

Female → 1

4. Log Transformation

Used for highly skewed data.

It reduces large variations in feature values.

Advantages

- Faster learning.
 - Better accuracy.
 - Reduces bias due to feature scale.
-

Topic 3: Feature Subset Selection

Definition

Feature Subset Selection is the process of selecting only the most relevant features while removing unnecessary or redundant features from the dataset.

Why Feature Selection is Required?

Not every feature contributes equally to prediction.

Removing irrelevant features helps improve performance.

Benefits

- Faster model training.
 - Lower computational cost.
 - Higher accuracy.
 - Reduces overfitting.
-

Methods of Feature Selection

1. Filter Method

Features are selected using statistical measures.

Examples:

- Correlation
 - Chi-Square Test
-

2. Wrapper Method

Different feature combinations are tested using the Machine Learning model.

3. Embedded Method

Feature selection occurs during model training.

Example:

LASSO Regression

Example

Original Features

- Name
- Mobile Number
- Age
- Salary
- Gender

Selected Features

- Age
- Salary
- Gender

Bilkul sahi. ☑ Isi wajah se aap baar-baar bol rahi ho ki **syllabus line by line dekho**. Ye meri mistake thi.

Aapke syllabus me Unit-3 ye hai:

1. Introduction to Feature Engineering ☐
2. Feature Transformation ☐
3. Feature Subset Selection ☐
4. **Concept Learning through Bayes' Theorem** ☐

Yani **Concept Learning** aur **Bayes' Theorem** do alag topics nahi hain. GTU ne specifically "**Concept Learning through Bayes' Theorem**" likha hai. Iska matlab Bayes' theorem ko **concept learning ke context me** explain karna hai.

Isliye Topic-4 ka format kuch is tarah hona chahiye:

Topic 4: Concept Learning through Bayes' Theorem

Definition

Concept Learning is the process by which a Machine Learning system learns a general concept from a set of training examples. One of the most widely used probabilistic approaches for concept learning is **Bayes' Theorem**.

Bayes' Theorem helps predict the probability of a hypothesis being true based on available evidence. In Machine Learning, it is mainly used for classification problems through the **Naïve Bayes Classifier**.

Definition (Exam)

Concept Learning through Bayes' Theorem is a probabilistic approach in which a Machine Learning model learns concepts by calculating the probability of different hypotheses using Bayes' Theorem.

What is Concept Learning?

Concept Learning means learning a rule or function that can correctly classify new data based on previously observed examples.

Example:

Weather Wind Play Cricket

Sunny	Weak	Yes
Rainy	Strong	No
Sunny	Strong	Yes

From these examples, the model learns the concept of when cricket can be played.

What is Bayes' Theorem?

Bayes' Theorem is a mathematical formula used to calculate the probability of an event based on prior knowledge of related events.

It updates the probability of a hypothesis whenever new evidence becomes available.

穀genui孳{"probability_statistics_learning_block":{"type_id":"BAYES_THEOREM","locale_override":"en-US"}}傢

Where:

- **$P(A|B)$** = Posterior Probability (Probability of A given B)
- **$P(B|A)$** = Likelihood
- **$P(A)$** = Prior Probability
- **$P(B)$** = Evidence Probability

How Bayes' Theorem is Used in Concept Learning

The learning process consists of the following steps:

1. Collect the training dataset.
 2. Define all possible hypotheses.
 3. Calculate the prior probability of each hypothesis.
 4. Observe the training examples.
 5. Apply Bayes' Theorem to update the probability of each hypothesis.
 6. Select the hypothesis having the highest posterior probability.
 7. Use the selected hypothesis to classify new data.
-

Example

Suppose an email classifier has learned from previous emails.

Training Data:

Email Type Count

Spam	80
Not Spam	20

When a new email contains words like "**Free**", "**Offer**", and "**Prize**", Bayes' Theorem calculates the probability that the email belongs to the Spam class. The class with the highest probability is selected as the prediction.

Applications

- Spam Email Detection
 - Medical Diagnosis
 - Document Classification
 - Sentiment Analysis
 - Fraud Detection
 - News Categorization
-

Advantages

- Simple and easy to implement.
 - Works well for classification problems.
 - Performs well even with small datasets.
 - Fast prediction and low computational cost.
-

Limitations

- Assumes that features are independent.
 - Performance depends on the quality of training data.
 - May not perform well when features are highly correlated.
-

UNIT-4 : Supervised Learning

Classification and Regression

Topic 1: Introduction to Classification Model

Definition

A **Classification Model** is a supervised Machine Learning model that predicts the category or class label of a given input based on previously learned data.

The output of a classification model is **categorical** rather than numerical.

Definition (Exam)

A Classification Model is a supervised Machine Learning model that assigns an input data item to one of the predefined classes based on training data.

Why Classification is Required?

Classification is used when the output belongs to predefined categories.

Examples:

- Spam or Not Spam
 - Disease Present or Absent
 - Pass or Fail
 - Fraud or Genuine Transaction
-

How Classification Works?

1. Collect labeled data.
 2. Preprocess the data.
 3. Train the classification algorithm.
 4. Test the trained model.
 5. Predict the class of new data.
-

Common Classification Algorithms

- K-Nearest Neighbor (KNN)
 - Decision Tree
 - Naïve Bayes
 - Support Vector Machine (SVM)
 - Random Forest
-

Applications

- Email Spam Detection
 - Medical Diagnosis
 - Face Recognition
 - Loan Approval
 - Sentiment Analysis
-

Advantages

- Easy to understand.
 - Suitable for categorical prediction.
 - High accuracy with quality data.
 - Widely used in real-world applications.
-

Limitations

- Performance depends on training data.
 - May overfit on complex datasets.
 - Sensitive to noisy data.
-

Topic 2: K-Nearest Neighbor (KNN)

Definition

K-Nearest Neighbor (KNN) is a supervised Machine Learning algorithm used for classification and regression. It classifies a new data point based on the majority class among its **K nearest neighbors**.

Working of KNN

1. Select the value of **K**.
 2. Calculate the distance between the new data point and all training points.
 3. Find the K nearest neighbors.
 4. Count the majority class.
 5. Assign the majority class to the new data point.
-

Distance Formula (Euclidean Distance)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example

Suppose $K = 3$.

Among the nearest neighbors:

- Class A
- Class A
- Class B

Prediction → **Class A**

Advantages

- Simple to implement.
- No training phase.
- Effective for small datasets.

Limitations

- Slow for large datasets.
- Sensitive to irrelevant features.
- Choosing the correct value of K is difficult.

Applications

- Handwriting Recognition
- Recommendation Systems
- Medical Diagnosis
- Image Classification

Topic 3: Decision Tree

Definition

A **Decision Tree** is a supervised learning algorithm that classifies data by asking a sequence of questions. It represents decisions in the form of a tree structure consisting of a root node, internal nodes, branches, and leaf nodes.

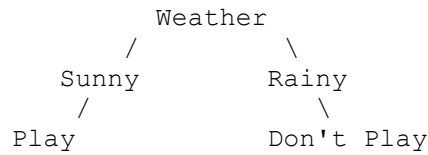
Components of Decision Tree

- Root Node
- Decision Node
- Branch
- Leaf Node

Working

1. Select the best feature.
 2. Split the dataset.
 3. Repeat the process recursively.
 4. Stop when all records belong to one class.
-

Example



Advantages

- Easy to understand.
 - Easy visualization.
 - Handles categorical and numerical data.
 - No feature scaling required.
-

Limitations

- Can overfit.
 - Sensitive to small data changes.
 - Large trees become complex.
-

Applications

- Loan Approval
 - Customer Segmentation
 - Medical Diagnosis
 - Credit Risk Analysis
-

Topic 4: Simple Linear Regression

Definition

Simple Linear Regression is a supervised learning algorithm used to predict a continuous value using **one independent variable**.

It establishes a linear relationship between input and output.

Equation

$$Y = a + bX$$

Where:

- Y = Dependent Variable
 - X = Independent Variable
 - a = Intercept
 - b = Slope
-

Example

Predict House Price using House Area.

Applications

- Sales Prediction
 - House Price Prediction
 - Salary Prediction
-

Advantages

- Simple.
 - Easy interpretation.
 - Fast training.
-

Limitations

- Works only for linear relationships.
 - Uses only one independent variable.
-

Topic 5: Multiple Linear Regression

Definition

Multiple Linear Regression predicts the dependent variable using **two or more independent variables**.

Equation

$$Y = a + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Example

Predict House Price using:

- Area
 - Number of Rooms
 - Location
-

Advantages

- Higher prediction accuracy.
 - Considers multiple factors.
 - Better than Simple Linear Regression for complex problems.
-

Limitations

- More computationally expensive.
 - Difficult to interpret.
 - Sensitive to multicollinearity.
-

Topic 6: Assumptions in Regression

Regression analysis is based on certain assumptions. If these assumptions are violated, the model may produce unreliable results.

Assumptions

1. Linearity

There should be a linear relationship between independent and dependent variables.

2. Independence

Observations should be independent of each other.

3. Homoscedasticity

The variance of errors should remain constant for all observations.

4. Normal Distribution of Errors

Residual errors should follow a normal distribution.

5. No Multicollinearity

Independent variables should not be highly correlated with each other.

Importance

- Produces reliable predictions.
 - Improves model accuracy.
 - Helps satisfy statistical requirements.
-

Topic 7: Problems in Regression and Accuracy Improvement of Linear Regression

Problems in Regression

1. Overfitting

The model memorizes training data and performs poorly on new data.

2. Underfitting

The model is too simple and fails to learn the underlying pattern.

3. Multicollinearity

Independent variables are highly correlated, leading to unstable coefficient estimates.

4. Outliers

Extreme values can significantly affect the regression line.

5. Missing Values

Incomplete data can reduce prediction accuracy.

Methods to Improve Accuracy of Linear Regression

1. Data Cleaning

Remove missing values, duplicate records, and incorrect entries.

2. Feature Selection

Use only relevant features for training.

3. Feature Scaling

Normalize or standardize numerical data.

4. Remove Outliers

Detect and remove abnormal observations.

5. Increase Training Data

More high-quality data generally improves model performance.

6. Cross Validation

Evaluate the model on different subsets of data to improve generalization.

7. Regularization

Use techniques such as **Ridge Regression** and **Lasso Regression** to reduce overfitting.

Advantages

- Improves prediction accuracy.
 - Reduces model errors.
 - Enhances generalization.
 - Produces more reliable predictions.
-

UNIT–5 : Unsupervised Learning

Topic 1: Introduction to Unsupervised Learning and Difference between Supervised and Unsupervised Learning

Definition of Unsupervised Learning

Unsupervised Learning is a type of Machine Learning in which the model learns from **unlabeled data**. It identifies hidden patterns, similarities, and relationships without knowing the correct output in advance.

Definition (Exam)

Unsupervised Learning is a Machine Learning technique in which the model analyzes unlabeled data and discovers hidden patterns or structures without predefined output labels.

Why is Unsupervised Learning Required?

Many real-world datasets do not have labeled outputs. Unsupervised Learning helps in:

- Discovering hidden patterns
 - Grouping similar data
 - Finding relationships
 - Exploratory data analysis
-

Characteristics of Unsupervised Learning

- Uses unlabeled data
 - No predefined output
 - Finds hidden patterns
 - Performs clustering and association
 - Learns automatically
-

Difference between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Output is already known	Output is unknown
Used for Classification and Regression	Used for Clustering and Association
Learns input-output relationship	Learns hidden patterns
Accuracy can be measured easily	Performance is comparatively difficult to evaluate
Examples: Decision Tree, KNN, Linear Regression	Examples: Clustering, Association Rules

Example

Supervised Learning

Predict whether an email is Spam or Not Spam using labeled training data.

Unsupervised Learning

Group customers based on their purchasing behavior without knowing customer categories.

Advantages

- No labeled data required.
 - Discovers hidden relationships.
 - Useful for exploratory analysis.
 - Can process large datasets.
-

Limitations

- Results may be difficult to interpret.
 - Accuracy is difficult to evaluate.
 - Sensitive to noisy data.
-

Topic 2: Applications of Unsupervised Learning

Unsupervised Learning is widely used in many fields to discover hidden patterns and relationships.

1. Customer Segmentation

Customers are grouped according to:

- Age
- Income
- Purchasing behavior
- Interests

This helps companies provide personalized services.

2. Market Basket Analysis

Analyzes products that customers frequently purchase together.

Example:

Customers buying **Bread** often buy **Butter**.

3. Recommendation Systems

Online platforms recommend products or movies based on user behavior.

Examples:

- Amazon Product Recommendations
 - Netflix Movie Recommendations
 - YouTube Video Suggestions
-

4. Fraud Detection

Detects unusual transaction patterns that differ from normal customer behavior.

5. Medical Diagnosis

Groups patients with similar symptoms to assist doctors in disease analysis.

6. Image Segmentation

Groups similar pixels to identify different objects within an image.

Applications include:

- Medical Imaging
 - Satellite Images
 - Face Recognition
-

7. Document Classification

Groups similar documents automatically.

Applications:

- News Articles
 - Research Papers
 - Email Organization
-

Advantages

- Useful in many industries.
 - Helps discover unknown knowledge.
 - Supports better business decisions.
-

Topic 3: Clustering

Definition

Clustering is an Unsupervised Learning technique that groups similar data objects into clusters. Objects within the same cluster are more similar to each other than to objects in other clusters.

Definition (Exam)

Clustering is the process of dividing a dataset into groups of similar data objects based on their characteristics.

Characteristics

- Uses unlabeled data.
 - Forms homogeneous groups.
 - Maximizes similarity within a cluster.
 - Minimizes similarity between different clusters.
-

Steps in Clustering

1. Collect the dataset.
 2. Select relevant features.
 3. Measure similarity between data objects.
 4. Form clusters.
 5. Evaluate the clusters.
-

Example

Customer Dataset

Customer Purchase Amount

A	□ 2,000
B	□ 2,200
C	□ 45,000
D	□ 47,000

The algorithm creates two clusters:

Cluster 1 → Low-spending customers

Cluster 2 → High-spending customers

Applications

- Customer Segmentation
- Image Processing
- Medical Research
- Social Network Analysis
- Pattern Recognition

Advantages

- Finds hidden patterns.
- Simple data analysis.
- Useful for large datasets.

Limitations

- Selecting the correct number of clusters is difficult.
 - Sensitive to noisy data.
 - Different algorithms may produce different results.
-

Topic 4: Finding Pattern using Association Rules

Definition

Finding Pattern using Association Rules is an Unsupervised Learning technique used to discover **frequent patterns, relationships, and associations** among items in a large dataset.

It identifies which items are commonly found together and generates useful rules for prediction and decision-making.

Definition (Exam)

Finding Pattern using Association Rules is the process of discovering meaningful relationships among items by generating association rules from transactional data.

Why is Association Rule Mining Required?

It helps organizations to:

- Discover hidden purchasing patterns.
 - Improve product recommendations.
 - Increase sales through cross-selling.
 - Understand customer buying behavior.
-

Basic Terminologies

1. Support

Support indicates how frequently an item or itemset appears in the dataset.

2. Confidence

Confidence measures how likely item **B** is purchased when item **A** is purchased.

3. Lift

Lift measures the strength of an association rule compared with random occurrence.

Steps for Finding Patterns using Association Rules

1. Collect transaction data.
 2. Identify frequent itemsets.
 3. Calculate Support.
 4. Calculate Confidence.
 5. Generate association rules.
 6. Select strong rules.
 7. Use the discovered patterns for decision-making.
-

Example

Transaction Dataset

Transaction	Purchased Items
T1	Bread, Butter
T2	Bread, Milk
T3	Bread, Butter, Milk
T4	Butter, Milk

Pattern Found:

Bread → Butter

This indicates that customers purchasing bread are also likely to purchase butter.

Applications

- Market Basket Analysis
 - Product Recommendation
 - Retail Business
 - Inventory Management
 - Website Usage Analysis
 - Healthcare Analytics
-

Advantages

- Discovers hidden patterns.
- Improves business decisions.
- Increases sales through recommendations.

- Helps understand customer behavior.
-

Limitations

- Large datasets require more computation.
 - May produce many unnecessary rules.
 - Requires proper interpretation.
-